
a Who is in each group, clustered at training
PAM on the training change in module eigengenes; gap selects K = 1, K = 2 used
C1 n = 7, Jaccard 0.67; C2 n = 7, Jaccard 0.56
PC2 (31%)
PC1 (34%)
C1 C2
HR LR
b Proteins per contrast
1,900 tested; 162 at FDR < 0.05
C1 (circ.)
C2 (circ.)
C1 (caut.)
C2 (caut.)
T1 (caut.)
T2 (caut.)
T3
Training (circ.)
Acute (caut.)
proteins (down | up)
Down Π < 0.05
Up p < 0.05
Down p < 0.05
Up Π < 0.05
c Proteome
42 biopsies, 1,900 proteins
PERMANOVA
time R² 0.07, p = 0.0006
group R² 0.06, p = 0.77
PC2 (9.5%)
PC1 (14.2%)
Phenotype
27 biopsies, six size and strength traits
PERMANOVA
time R² 0.04, p = 0.037
PC2 (21.4%)
PC1 (62.3%)
C1 C2
T1 T2 T3
d Pathways shared within each window
fgsea FDR < 0.05 in both groups of a window (interaction: either, top 5), shown in all nine contrasts; * opposite signs
Training
Acute
C1 - C2
Interaction
Oxidative phosphorylation
Nonsense mediated decay NMD
Striated muscle contraction
RNA splicing via transesterification reac...
Protein DNA complex organization
Sarcomere organization
Organic acid catabolic process
Sars cov 1 infection
Fatty acid metabolism
Generation of precursor metabolites and e...
MYC targets v1
Sulfur compound metabolic process
Activation of the mRNA upon binding of th...*
mRNA splicing
Mapk family signaling cascades
Heterochromatin organization
Cytoplasmic translation
Monosaccharide catabolic process
C1 C2 C1 C2 T1 T2 T3 Training Acute
NES
-log10 FDR
Database
Hallmark
Reactome
GO BP
e Training signature rank
fry: each group's p < 0.05 training sets in the other group's ranking
C2 training sets in C1's ranking
up (n = 83), 41% same way, FDR 0.034
down (n = 78), 36% same way, FDR 0.048
C1 training sets in C2's ranking
up (n = 243), 45% same way, FDR 0.84
down (n = 164), 20% same way, FDR 0.29
f Acute signature rank
fry: each group's p < 0.05 acute sets in the other group's ranking
C2 acute sets in C1's ranking
up (n = 130), 75% same way, FDR 0.13
down (n = 102), 75% same way, FDR 0.085
C1 acute sets in C2's ranking
up (n = 55), 65% same way, FDR 0.41
down (n = 116), 74% same way, FDR 0.31
rank in the other group's training contrast, up to down (n = 1900)
rank in the other group's acute contrast, up to down (n = 1900)